Proposed Solution

Date	23 October 2026
Team ID	0827CY231029
Project Name	Waypoint - AI Travel Itinerary Planner
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

Waypoint is an AI-assisted travel itinerary planner for solo travelers, couples and small groups. It turns destination and traveler preferences into a structured daily plan, while offering curated templates for a quick start.

Project Overview

Field	Details
Objective	Reduce manual trip-planning effort and help travelers review a clear, personalized daily schedule.
Scope	Browser-based destination discovery, curated itineraries, custom generation, saved preferences, trip management, public sharing and exports.

Problem Statement

Field	Details
Description	Travel information is scattered across websites, maps and messages. Generic lists do not combine personal interests, pace, costs and daily logistics.
Impact	A single saved itinerary improves planning clarity, makes changes easier and gives companions a common plan to view.

Proposed Solution

Field	Details
Approach	Collect destination, dates, budget, travelers, style, pace and notes; build an AI prompt; use local rules if no model succeeds; store the resulting plan.
Key Features	Six curated destinations; custom AI builder; authentication; preferences; daily timeline; saved-trip edits; destination recommendations; share links; JSON/Markdown/ICS export.

Proposed Solution (continued)

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Development computer and browser	Proposed baseline: 8 GB RAM and available disk space for dependencies. Hosted AI does not require a local GPU.
Software	Languages, runtime and frameworks	Python 3.11+; Node.js 18+; FastAPI; React/Vite; dependency manifests supplied.
Data	Relational storage and curated content	Supabase Postgres when configured; SQLite locally; destination catalog and curated itinerary data in source files.
AI service	Optional model access	Groq/OpenAI/Hugging Face credentials or explicitly selected Ollama. Local rules support operation without keys.
Development environment	Editor, version control and debugging	VS Code or another editor; Git/GitHub; browser dev tools; FastAPI /docs.
Hosting	Production deployment configuration	Render API, Vercel frontend and Supabase database; public URLs require deployment.

Expected Outcomes and Boundaries

The solution produces an editable planning aid with estimated logistics and costs. Booking, payments, guaranteed venue availability, real-time routing and multi-city collaboration are outside the current scope.

Project sources: README.md; 7. Project Documentation/SETUP.md; backend catalog, AI engine, repository and frontend pages.